

# Dinesh Kothandaraman

📍 West Haven, CT ✉ dineshkothand@gmail.com ☎ +1 475-227-5994 [in Dinesh Kothandaraman](#)

🌐 Dinesh-Kothandaraman

## Summary

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Data Science graduate student with 5.6 years of professional experience, strong expertise in Python, SQL, machine learning, deep learning, and data analysis. Proven ability to apply technical skills and project experience to solve real-world problems in dynamic, data-driven environments.

## Education

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| <b>MS</b> | <b>University of New Haven, Data Science</b>                                                                                                                                                                                                                                                     | Aug 2023 – May 2025 |
|           | <ul style="list-style-type: none"><li>• GPA: 3.9/4.0</li><li>• <b>Coursework:</b> Artificial Intelligence, Machine Learning, Distributed and Scalable Data Engineering, Programming for Data Science, Deep Learning, Natural Language Processing, Power BI and Bayesian Data Analysis.</li></ul> |                     |
| <b>BE</b> | <b>Chennai Institute of Technology , Mechatronics Engineering</b>                                                                                                                                                                                                                                | Jun 2013 – Apr 2017 |
|           | <ul style="list-style-type: none"><li>• GPA: 7.35/10</li><li>• <b>Coursework:</b> Computer Programming, Transforms and Partial Differential Equations, Statistics and Numerical Methods, Object Oriented Programming in C++.</li></ul>                                                           |                     |

## Technical Skills

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**Programming Languages:** Python, SQL, JavaScript, HTML, COBOL, JCL, VSAM, CICS

**Tools & Databases:** AWS, Visual Studio Code, Jupyter Notebook, SQLiteStudio, MySQL, SQL Server

**Libraries:** NumPy, Scikit-learn, Pandas, Matplotlib, PostgreSQL, Psycopg2, SymPy, PyTorch

**Certifications:** Python Programming, Fundamentals of Data Analytics with AWS, HarvardX - CS50's Introduction to Computer Science, LOMA 281

## Experience

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| <b>Cognizant Technology Solutions India Private Ltd, Mainframe Developer</b>                                                                                                                                                                                                                                                                                                                                                                                                | Chennai, India       |
| <ul style="list-style-type: none"><li>• Designed and developed solutions using COBOL, JCL, VSAM, DB2, and JavaScript based on client requirements.</li><li>• Authored and optimized SQL queries for data extraction and performance tuning.</li><li>• Automated repetitive tasks with custom tools developed in REXX, significantly improving efficiency.</li><li>• Adapted seamlessly between Agile and Waterfall methodologies to meet varying project demands.</li></ul> | Jan 2021 – July 2023 |
| <b>Genius Consultants Ltd, Quality Inspector</b>                                                                                                                                                                                                                                                                                                                                                                                                                            | Chennai, India       |
| <ul style="list-style-type: none"><li>• Conducted periodic inspections of headlamps and rear lamps to ensure compliance with quality standards.</li><li>• Performed final visual inspections, guaranteeing high product standards before shipment.</li></ul>                                                                                                                                                                                                                | Apr 2019 – May 2020  |
| <b>Rytik's Auto Pvt. Ltd, Quality Engineer</b>                                                                                                                                                                                                                                                                                                                                                                                                                              | Chennai, India       |
| <ul style="list-style-type: none"><li>• Led periodic line inspections of armature shafts, ensuring adherence to product specifications.</li></ul>                                                                                                                                                                                                                                                                                                                           | Mar 2018 – Mar 2019  |

- Authored and implemented First Off Approval (FOA) and Standard Operating Procedures (SOPs) for products.

## Publication

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### Hybrid CNN-LSTM Model for Jail Activity Recognition

Oct 2024

Published in 2024 IEEE 15th Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON)

<https://ieeexplore.ieee.org/abstract/document/10754670>

## Projects

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### Gender and Age Detection

- Developed a Python-based deep learning project to detect faces and classify age and gender from images.
- Leveraged pre-trained models to achieve high accuracy and real-time performance.
- Utilized advanced computer vision techniques for optimized analysis.

### Remote Patient Care with AWS and IoT

- Designed a scalable health monitoring system integrating IoT devices with AWS tools.
- Used **AWS Glue** for processing health data and Python for anomaly detection in real-time metrics.
- Developed dashboards in **AWS QuickSight** to visualize and monitor health indicators.

### Jail Activity Recognition Using CNN and LSTM

- Built a real-time activity recognition model combining CNN and LSTM architectures.
- Processed and analyzed time-series data, achieving high model accuracy.

### Text Summarization - Natural Language Processing

- Developed three text summarization models: an LSTM with attention mechanism and two pre-trained transformer models (T5 and BART), focusing on both abstractive and extractive techniques.
- Fine-tuned T5 and BART models on domain-specific datasets, leveraging their encoder-decoder architecture for high-quality summarization with improved ROUGE and BLEU scores.
- Implemented an LSTM with attention to enhance focus on relevant text segments, visualizing attention weights for interpretability and improving model performance on custom datasets.
- Evaluated and compared models on benchmark datasets, demonstrating the superiority of transformer-based methods while showcasing expertise in both traditional and cutting-edge NLP techniques.

### Car Logo Detection - Deep Learning

- Developed a car logo detection model using PyTorch and transfer learning, leveraging pre-trained CNN architectures (ResNet, VGG) to achieve high accuracy on custom datasets.
- Implemented transfer learning by fine-tuning a pre-trained model on a labeled car logo dataset, reducing training time and improving generalization.
- Integrated ClearML for streamlined experiment tracking, performance monitoring, and efficient model versioning during the development lifecycle.
- Optimized model performance through data augmentation, learning rate scheduling, and loss function tuning, achieving significant improvement in precision and recall metrics.